1. Hospital Doctor Scheduling using Backtracking Search

A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

i. D1 cannot work Night shift

ii. D2 must work before D3 (Morning < Afternoon < Night)

iii. Only one doctor per shift

iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Problem

Assign 3 doctors (D1, D2, D3) to 3 shifts.

Shifts

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3.
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor gets exactly one shift.

Step 1: Initial Assignment

No doctor is assigned.

Doctor Shift

D1 -

D2 -

D3 -

Step 2: Assign D1

Try Morning.

D1 → Morning
Doctor Shift

D1 Morning

D2 -

D3 -

Constraint Check

● D1 not Night

● One doctor per shift

Step 3: Assign D2

Morning already occupied.

Try Afternoon.

D2 → Afternoon

Doctor Shift

D1 Morning

D2 Afternoon

D3 -

Constraint Check

● One doctor per shift

Step 4: Assign D3

Morning (Not allowed)

Afternoon (Already occupied)

Night

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

Constraint Check

● D3 not Morning

● D2 before D3 (Afternoon before

Night) ● One doctor per shift

● All assigned

No need to backtrack because all constraints are satisfied.

Search Tree

Start

|

D1 → Morning

|

D2 → Afternoon

|

D3 → Morning

|

D3 → Afternoon

|

D3 → Night

Final Schedule

Shift Doctor

Morning D1

Afternoon D2

Night D3

Result

A valid schedule is:

● D1 → Morning

● D2 → Afternoon

● D3 → Night

2. Robot Path Finding using BFS

Since the question does not provide the grid, assume the following 5 ×

5 grid. S . . X .

. X . X .

. . . . .

X X . X .
. . . . G

Legend

- S = Start
- G = Goal
- X = Obstacle
- . = Empty

Coordinates

S = (0,0)

G = (4,4)

Manhattan Distance

Formula |x1-x2| + |y1-y2|

Example

From Start

|0-4| + |0-4|

=8

BFS Step-by-Step

Step 1

Queue

[(0,0)]

Expand

(0,0)

Visited

(0,0)

Add neighbors

(1,0)

(0,1)

Queue

[(1,0),(0,1)]

Step 2

Expand
(1,0)

Add

(2,0)

Queue

[(0,1),(2,0)]

Step 3

Expand

(0,1)

Right

(0,2)

Queue

[(2,0),(0,2)]

Step 4

Expand

(2,0)

Add

(2,1)

Queue

[(0,2),(2,1)]

Step 5

Expand

(0,2)

Add

(1,2)

Queue

[(2,1),(1,2)]

Continu

e

Expand

(2,1)
↓

(2,2)

↓

(2,3)

↓

(2,4)

↓

(3,4)

↓

(4,4)

Goal Found.

BFS Expansion

Order (0,0)

↓

(1,0)

↓

(0,1)

↓

(2,0)

↓

(0,2)

↓

(2,1)

↓

(1,2)

↓

(2,2)

↓

(2,3)

↓

(2,4)

↓

(3,4)

↓

(4,4)

Optimal Path

S

↓

(1,0)

↓

(2,0)

↓

(2,1)

↓

(2,2)

↓

(2,3)

↓

(2,4)

↓

(3,4)

↓

G

Total Cost

Each move costs 1.

Number of moves = 8

Total Cost = 8

Result

Optimal Path

S → (1,0) → (2,0) → (2,1) →

(2,2) → (2,3) → (2,4) → (3,4)

→ G Total Cost = 8

2. Autonomous Rescue Robot

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

(a) Constraint Analysis

Constraint Effect on Robot

Move only Up, Down, Left, Right Robot cannot move diagonally.

Obstacles Robot must avoid blocked rooms.

Limited sensing Robot knows only nearby rooms and explores gradually. Move

cost = 1 Every step increases travel cost.

Risky zone cost = +2 Robot prefers safer paths even if slightly longer. Online update

Robot updates its map whenever it discovers new obstacles.

(b) Solution Approach (Uniform Cost Search + Online Search)

Step 1

Robot starts at S.

Step 2
Observe adjacent cells.

Step 3

Ignore blocked cells.

Step 4

Calculate total path cost.

Normal cell = 1

Risk cell = 3

(1 movement +2 risk)

Step 5

Choose the path with the lowest total cost using Uniform Cost Search

(UCS). Step 6

Move to the selected room.

Step 7

Sense new neighboring cells.

Step 8

If new obstacles are found, update the map and recalculate the

best path. Step 9

Repeat until reaching G.

Flow

Start

↓

Sense nearby cells

↓

Remove blocked paths

↓

Calculate costs

↓

Choose lowest-cost path

↓

Move

↓

Update map
↓

Repeat

↓

Reach Goal

(c) AI Concepts

1. Intelligent Agent

● The robot senses its environment.

● It thinks about the best move.

● It acts by moving to the next room.

Agent Cycle

Sense

↓

Think

↓

Act

2. Rationality

A rational robot always chooses the action that minimizes total cost while safely reaching the survivor. Example

Short risky path = Cost 12

Long safe path = Cost 9

Robot chooses safe path.

3. Environment Type

Property Type

Observability Partially Observable

Determinism Dynamic (new obstacles may appear)

Agent Single Agent

State Sequential

Environment Unknown initially

(d) Why Uniform Cost Search + Online Search?

Uniform Cost Search (UCS):

● Finds the lowest-cost path.

● Considers both movement and risk costs.

● Suitable when actions have different costs.

Online Search:

● Works well when the environment is partially observable. ● Allows the robot to update its knowledge dynamically as it explores. ● Adapts if new obstacles or risky zones are discovered.

Advantages

● Minimizes total cost.

● Avoids risky areas when possible.

● Handles unknown environments.

● Replans automatically after new information.